

Experiment 2 Report

Design and Performance Measurement of Negative Feedback Amplifiers in SPICE

Prepared from the provided experiment manual and simulation-result screenshots

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Experiment aim	Use SPICE to design negative-feedback amplifiers and evaluate gain, passband, and feedback effects.
Device and supply	LT1001 operational amplifiers powered by +12 V and -12 V supplies.
Main signals	1 kHz sine input; 1 V, 100 mV, and 200 mV cases; AC sweep from 1 Hz to 200 MHz.
Main conclusion	Negative feedback makes the gain predictable, improves stability, and broadens the useful passband at the cost of lower mid-band gain.

Abstract

This experiment investigated negative feedback in operational-amplifier circuits using SPICE simulation. A simple inverting amplifier was first simulated to verify that, with equal input and feedback resistors, the closed-loop voltage gain is approximately -1 and the output waveform is inverted. A two-stage LT1001 amplifier was then analyzed with and without the global feedback branch. The AC-response screenshots show that the feedback connection lowers the closed-loop gain but improves the usable passband and produces a more controlled frequency response. These results agree with the theoretical feedback relation $A_f = A/(1 + A\beta)$, where a large loop gain makes the closed-loop gain mainly determined by the feedback network.

1. Objectives

- Build negative-feedback amplifier circuits in SPICE using the LT1001 operational amplifier model.
- Verify the closed-loop gain of a basic inverting amplifier from transient waveforms.
- Measure and compare voltage gain and passband through AC analysis before and after connecting the feedback part.
- Observe how feedback affects gain, bandwidth, and output stability when the input amplitude is increased.

2. Theory

An ideal operational amplifier has very high open-loop differential gain, very high input resistance, and very low output resistance. Because the open-loop gain is large, a negative-feedback op-amp circuit can be analyzed using the virtual-short relation $U_+ \approx U_-$ and the virtual-open relation $I_+ \approx I_- \approx 0$. In a feedback amplifier, part of the output signal is returned through a feedback network and subtracted from the source input signal. For an open-loop amplifier gain A and feedback factor β , the closed-loop gain is

$$A_f = x_o / x_s = A / (1 + A\beta)$$

When the loop gain $A\beta$ is much larger than 1, the amplifier gain becomes approximately

$$A_f \approx 1 / \beta$$

Therefore, the closed-loop gain is mainly controlled by the passive feedback network rather than by the exact open-loop gain of the op-amp. For the basic inverting amplifier used in this experiment, the expected voltage gain is

$$A_v = V_o / V_s = -R_f / R_1 = -10 \text{ k}\Omega / 10 \text{ k}\Omega = -1$$

For AC analysis, the passband is evaluated with the -3 dB cutoff frequencies. The bandwidth is

$$BW = f_H - f_L$$

The AC source was set to a small-signal amplitude of 1 V, so the LTspice plot of $V(n008)$ in dB can be read directly as voltage gain in dB. The linear gain is calculated as

$$A_v(V/V) = 10^{(G_{dB} / 20)}$$

3. Simulation Setup

Item	Basic inverting test	100 mV AC-analysis test	200 mV comparison test
Op-amp	LT1001	Two LT1001 stages	Two LT1001 stages
Supply	+12 V and -12 V	+12 V and -12 V	+12 V and -12 V
Input	SINE(0 1 1K)	SINE(0 100mV 1K), AC 1	SINE(0 200m 1k), AC 1
Analysis	Transient, stop time 20 ms	AC sweep: .ac dec 101 1 200Meg	AC sweep: .ac dec 101 1 200Meg
Measured output	$V(n002)$ compared with input $V(n004)$	$V(n008)$	$V(n008)$

4. Procedure and Results

The procedure followed the instruction manual: build the feedback amplifier circuit, run a transient simulation to verify the gain, perform AC analysis, connect the feedback part, and compare the gain and passband. The numerical values below are graphical estimates read from the provided simulation screenshots.

Summary of estimated measurements from the provided screenshots:

Test	Feedback condition	Main gain estimate	Passband / response estimate	Interpretation
Basic inverting amplifier	Local inverting feedback, $R_f = R_1 = 10\text{ k}\Omega$	$A_v \approx -1$ (0 dB magnitude)	1 kHz transient; input and output have equal peak magnitude and 180° phase difference	Confirms the theoretical gain $-R_f/R_1$.
100 mV AC test	Global feedback branch initially open	$\approx 30\text{ dB}$, about 31.6 V/V	$f_L \approx \text{a few Hz}$; $f_H \approx 80\text{--}100\text{ kHz}$	High gain but narrower useful bandwidth.
100 mV AC test	Global feedback connected	$\approx 23\text{ dB}$, about 14 V/V	$f_L \approx \text{a few Hz}$; $f_H \approx 0.2\text{ MHz}$	Feedback reduces gain and increases bandwidth.
200 mV test	Feedback removed	Very high low-frequency gain, near open-loop behavior	No controlled flat passband; strong roll-off and phase shift	Large gain can lead to output saturation or distortion in time-domain operation.
200 mV test	Feedback connected	$\approx 27\text{ dB}$, about 22 V/V	Controlled passband with some high-frequency peaking and later roll-off	Feedback restores a usable, predictable response.

4.1 Basic inverting feedback amplifier

Figure 1 shows the verification circuit. The non-inverting input is grounded, $R_1 = 10\text{ k}\Omega$, and $R_f = 10\text{ k}\Omega$, so the feedback network fixes the closed-loop gain at -1. The input source is a 1 kHz, 1 V sine wave, and the circuit is simulated for 20 ms.

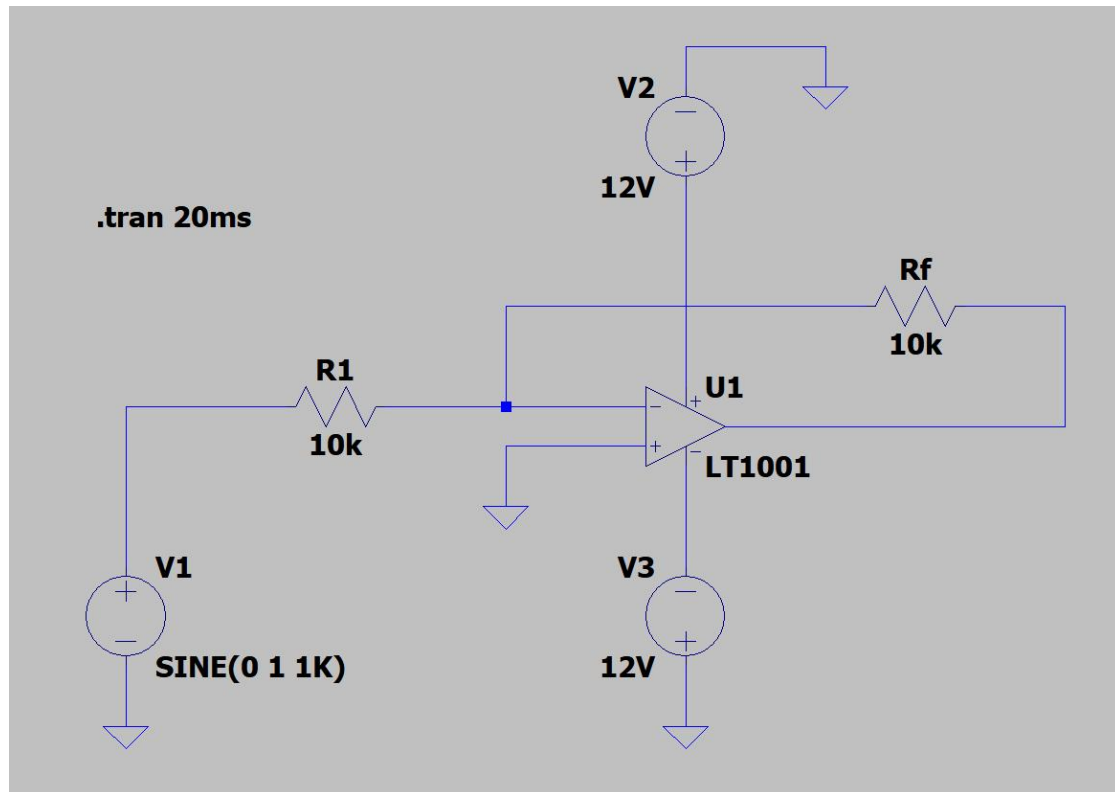


Figure 1. Basic LT1001 inverting negative-feedback amplifier for transient gain verification.

The transient result in Figure 2 shows two sinusoidal waveforms with nearly equal peak magnitude. The output is phase-inverted relative to the input, which verifies the expected voltage gain $A_v \approx -1$.

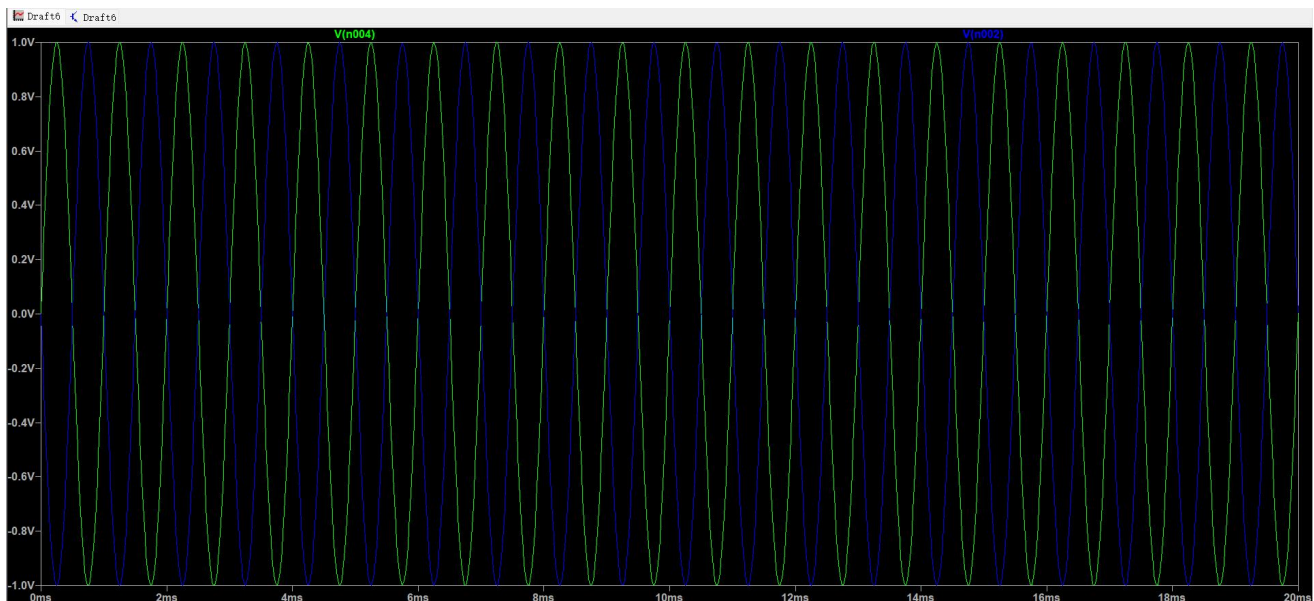


Figure 2. Transient simulation of the basic amplifier: equal amplitude and opposite phase verify $A_v \approx -1$.

4.2 AC analysis before connecting the feedback part

The two-stage amplifier for AC analysis is shown in Figure 3. In this screenshot, the global feedback branch containing R_9 and C_3 is not connected to the input summing node. The input source is set to SINE(0 100 mV 1 kHz) and AC 1, so the Bode plot directly indicates the small-signal gain in dB.

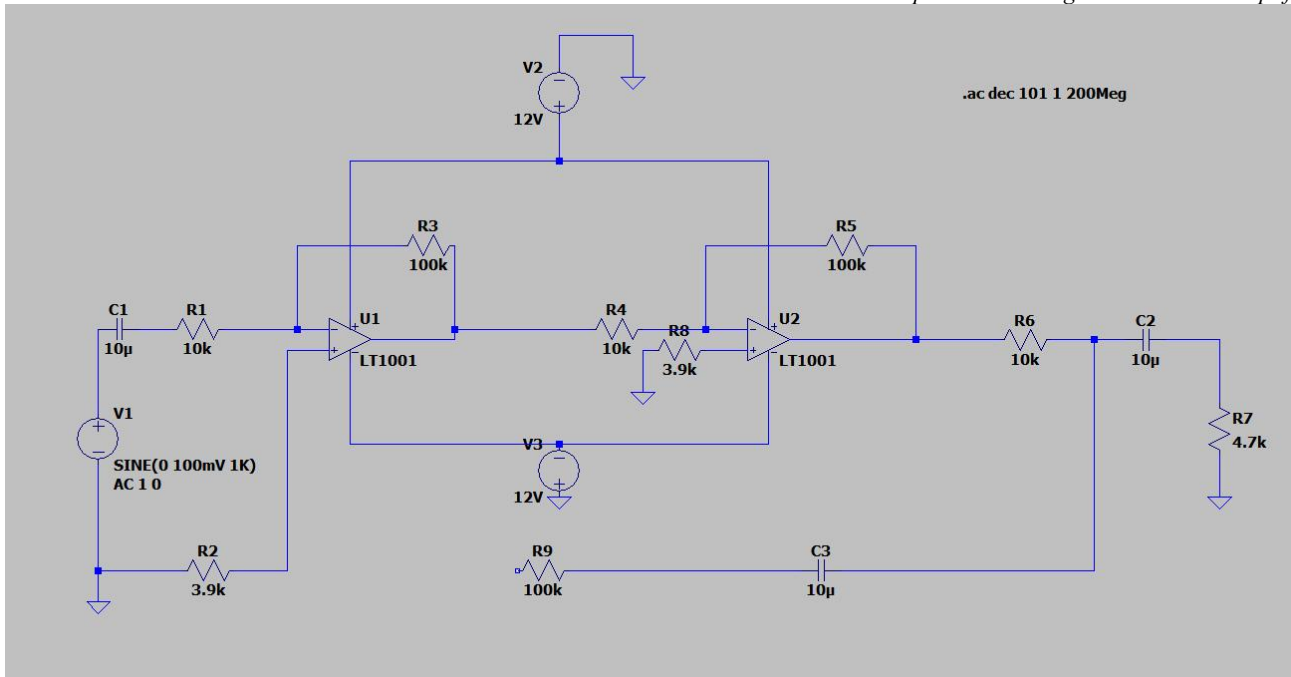


Figure 3. Two-stage LT1001 circuit with the global feedback branch open for the 100 mV AC-analysis case.

The full Bode plot in Figure 4 and the passband-scale plot in Figure 5 show a mid-band gain of about 30 dB, corresponding to approximately 31.6 V/V. The response is reasonably flat through the mid-band, but the high-frequency cutoff appears around 80-100 kHz from the -3 dB point.

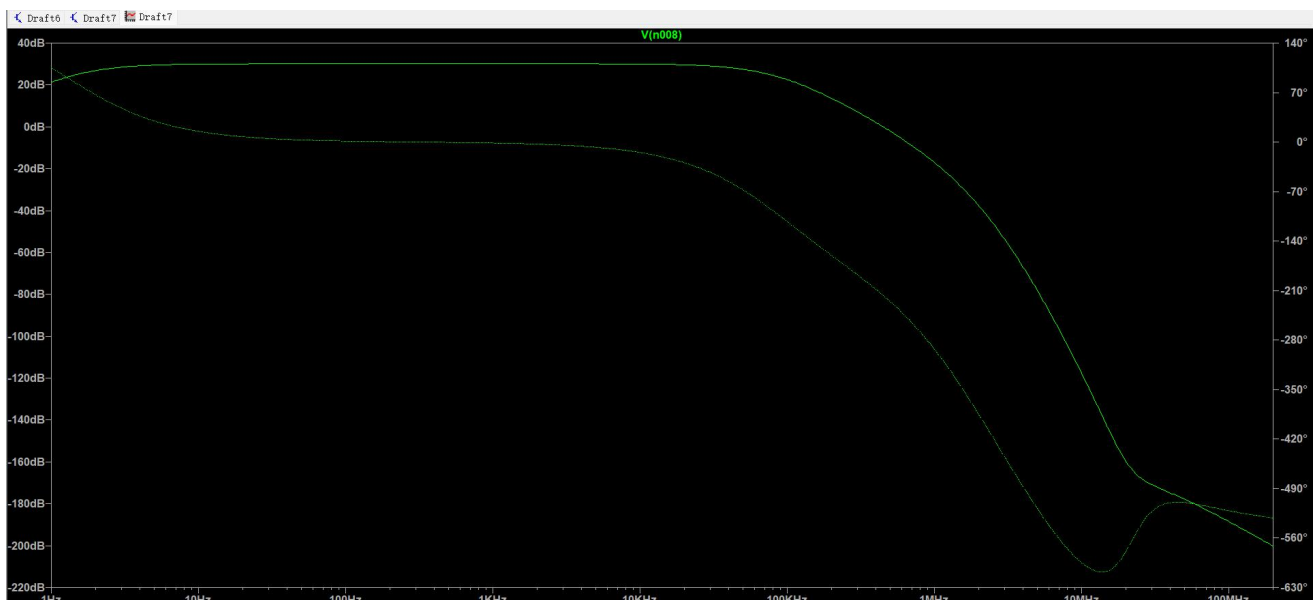


Figure 4. Full AC magnitude and phase response before connecting the global feedback branch.

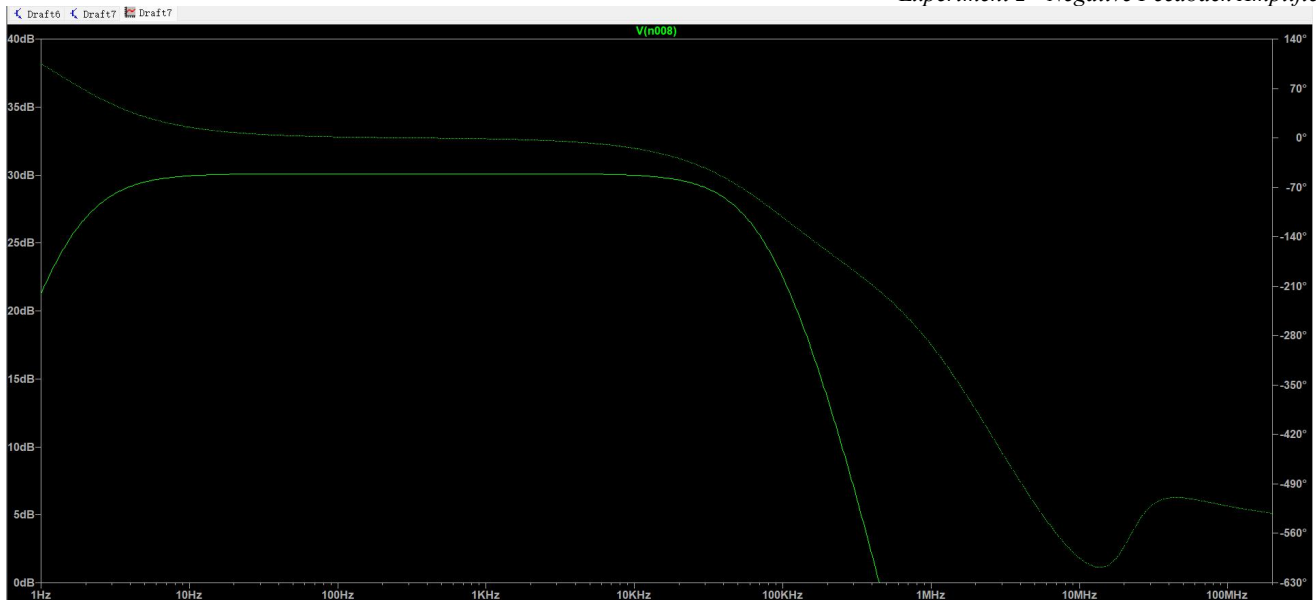


Figure 5. Passband-scale view used to estimate the cutoff frequencies before connecting feedback.

4.3 AC analysis after connecting the feedback part

Figure 6 shows the feedback part connected. The R9-C3 branch now returns output information to the first-stage input network, creating a global negative-feedback path. This changes the closed-loop gain and frequency response.

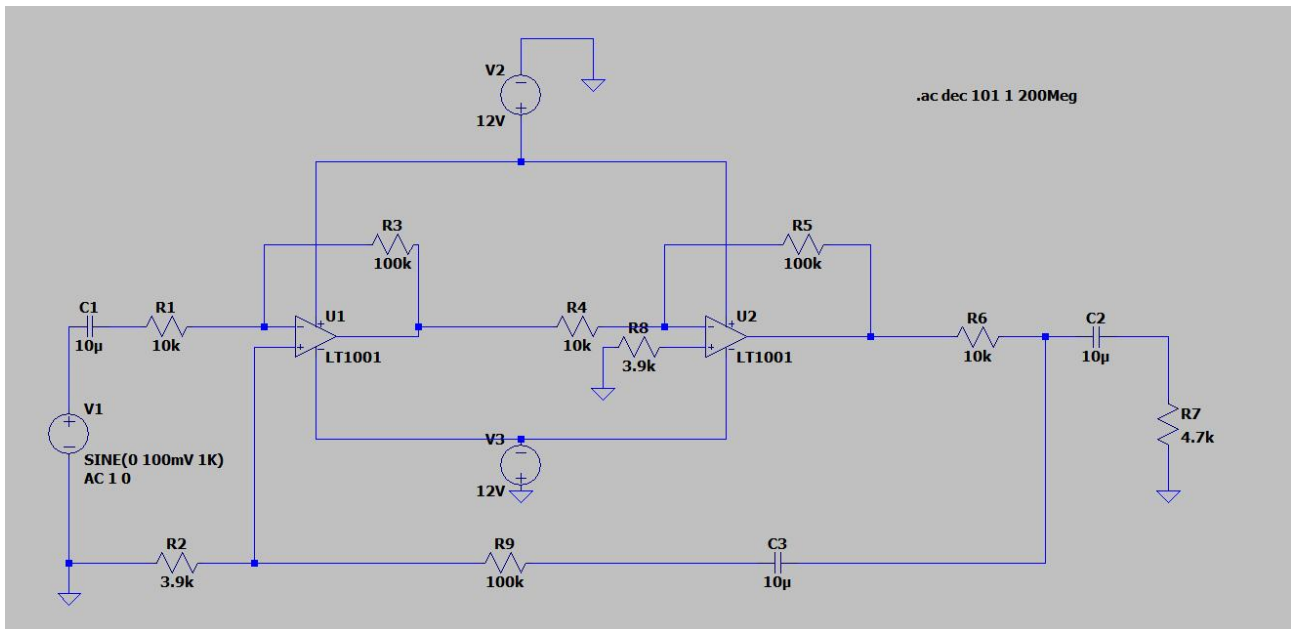


Figure 6. Two-stage LT1001 circuit after connecting the global feedback branch for the 100 mV test.

The response in Figure 7 shows that the mid-band gain is reduced to about 23 dB, or approximately 14 V/V. At the same time, the upper cutoff frequency is shifted upward to roughly 0.2 MHz. This illustrates the typical gain-bandwidth trade-off of negative feedback: gain decreases, while bandwidth and response control improve.

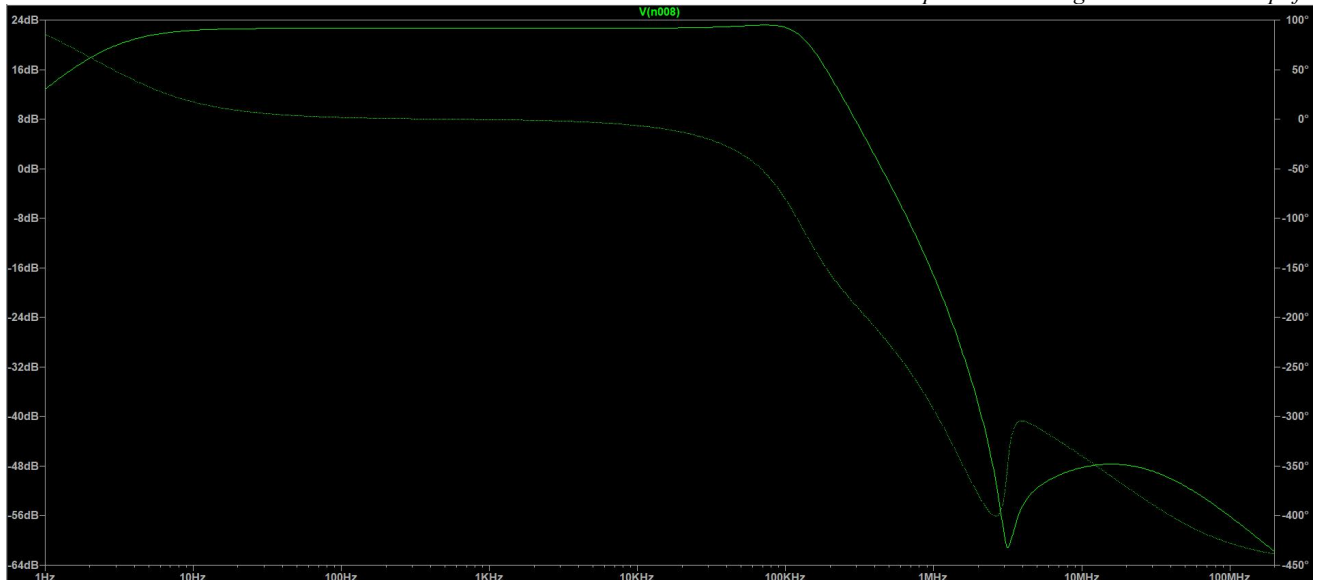


Figure 7. AC response after connecting feedback: lower mid-band gain and wider useful bandwidth.

4.4 Increased input amplitude: feedback removed and reconnected

The final comparison used a 1 kHz, 200 mV input. In Figure 8, the feedback branch containing R4 and C2 is open, so the circuit operates with far less closed-loop control.

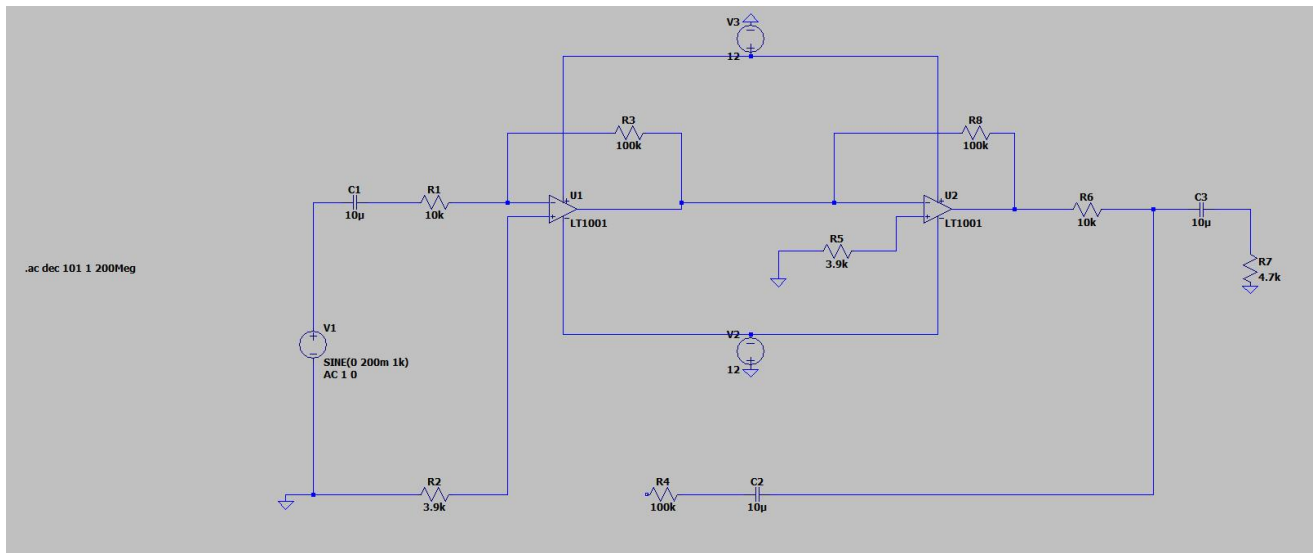


Figure 8. 200 mV case with the global feedback path open.

The corresponding response in Figure 9 has extremely high low-frequency gain and lacks a well-controlled flat passband. This behavior explains why a large input signal can easily force an amplifier without feedback into output limitation or distortion.

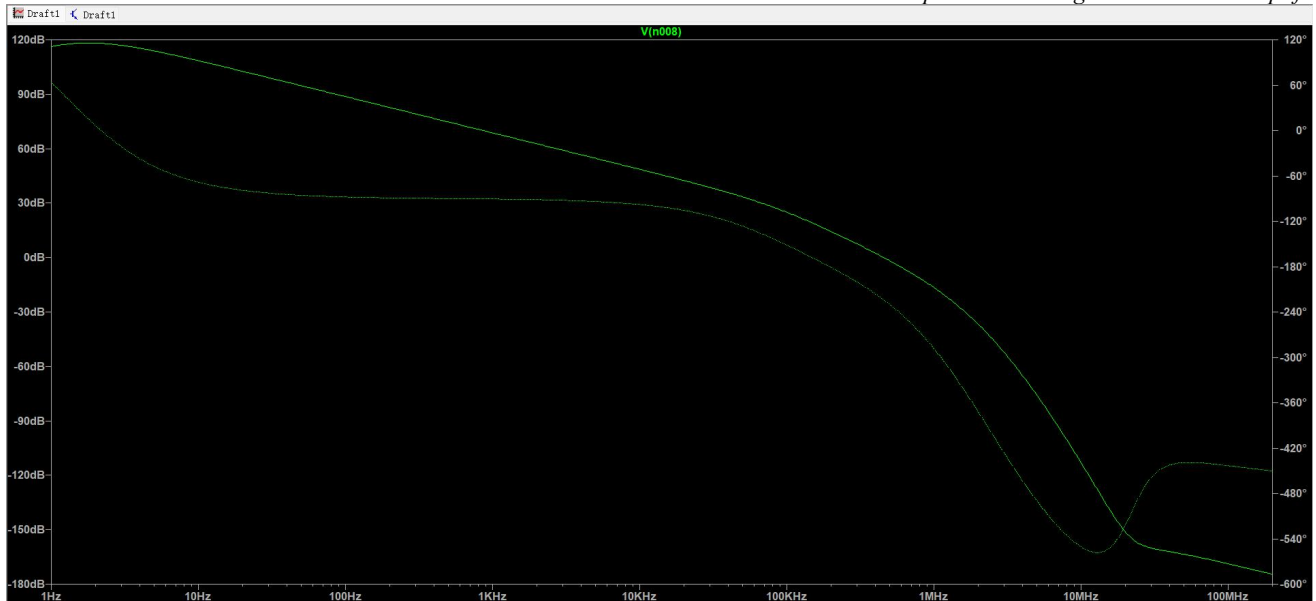


Figure 9. AC response with feedback removed in the 200 mV case: high gain and poor closed-loop control.

In Figure 10, the feedback branch is connected again. The output response in Figure 11 becomes much more controlled, with an estimated mid-band gain of about 27 dB. Although some high-frequency peaking is visible, the response is far more useful than the no-feedback case.

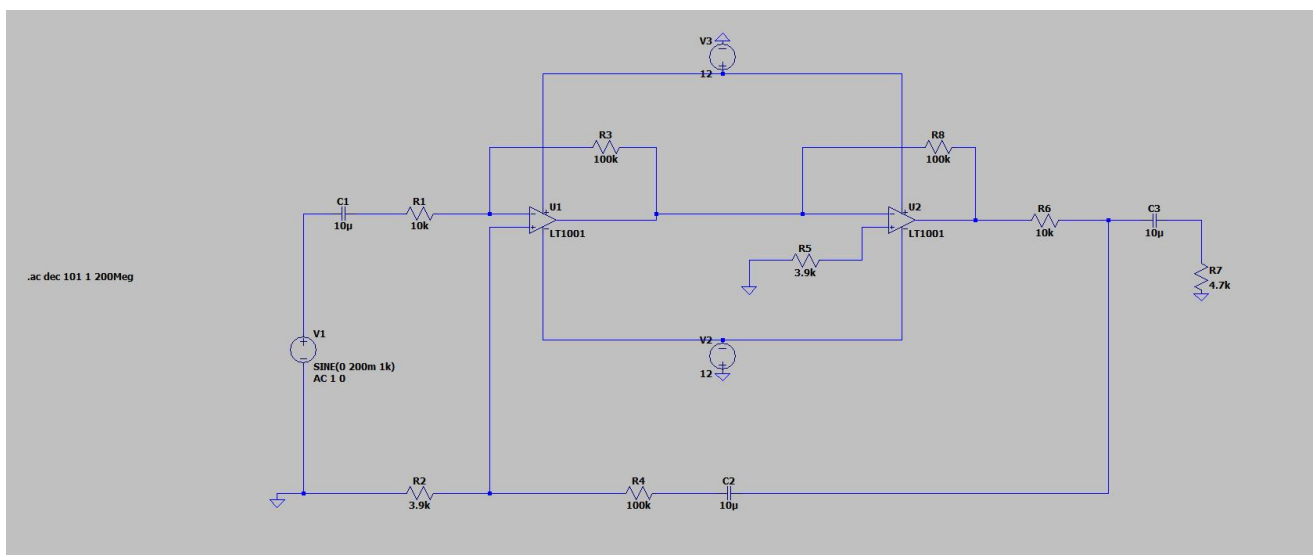


Figure 10. 200 mV case after reconnecting the global feedback path.



Figure 11. AC response after feedback is reconnected in the 200 mV case.

5. Discussion

The transient simulation of the basic inverting amplifier confirms the virtual-ground analysis. Because R_f and R_1 are equal, the output magnitude matches the input magnitude, and the negative sign appears as a 180° phase reversal. The two-stage AC simulations show the practical effect of global feedback. Before feedback is connected, the amplifier has a higher mid-band gain but the response is less controlled and the bandwidth is smaller. After feedback is connected, the gain decreases but the upper cutoff frequency increases, giving a wider passband. This is consistent with the feedback equation $A_f = A/(1 + A\beta)$: increasing feedback reduces the dependence on open-loop gain and improves the closed-loop response.

The 200 mV comparison further demonstrates why feedback is important. With the feedback path open, the gain is very high and the response does not have a useful, predictable flat region. In a time-domain test this condition would be more likely to produce output clipping or distortion because the op-amp is asked to amplify a larger input without sufficient feedback control. Reconnecting feedback stabilizes the gain and makes the output response usable again.

The main uncertainty in the numerical values is that the figures were screenshots rather than exported waveform data. Therefore, the cutoff frequencies and gains were read graphically from the plot axes. For exact final values, LTspice cursors should be used at the -3 dB points and at the chosen mid-band frequency.

6. Conclusion

- The basic inverting amplifier produced the expected closed-loop gain of approximately -1, with the output inverted relative to the input.
- The AC-analysis circuits demonstrated the gain-bandwidth trade-off: connecting the feedback path reduced mid-band gain but increased the useful passband.
- Negative feedback made the amplifier response more predictable and less dependent on the op-amp open-loop gain.
- When the input amplitude was increased to 200 mV, the no-feedback condition showed excessive gain and poor control, while the feedback condition restored a usable response.